Where and when a neighbour cell first reaches thermal-runaway onset temperatures: resolved against lumped, a pre-registered two-cell LG M50 pilot

Siddharth Satte, Independent researcher, Pune, India, 7 October 2026. Predictions commit c57ccb8, public at 06:42 UTC before any coupled resolved run. doi.org/10.5281/zenodo.23253738; github.com/siddharth161204-wq/m50-two-cell-pilot.

1. With copper busbars (500 mW/K in all) and the largest Databank energy (33.2 kJ), the resolved neighbour's hottest point first reaches 183.5 C, the mean T_in of Koenig, Zhao and Deng, 64 s before the lumped node: 5.7 s against 69.6 s for a 10 s release, 15.5 s against 79.2 s for 30 s. The hottest 1 % follows at 11.6 s and 25.0 s, but the resolved mean never reaches 183.5 C: the lumped node is a minute late for the hottest point and 44 K too hot for the mean (30 K two-way).
2. The smallest busbar conductance at which only the resolved model reaches 183.5 C is 200 mW/K (100 mW/K for the band's lower edge, 169.0 C); in each of those cases at most 0.52 % of the cell, a third of a gram, gets above 183.5 C.

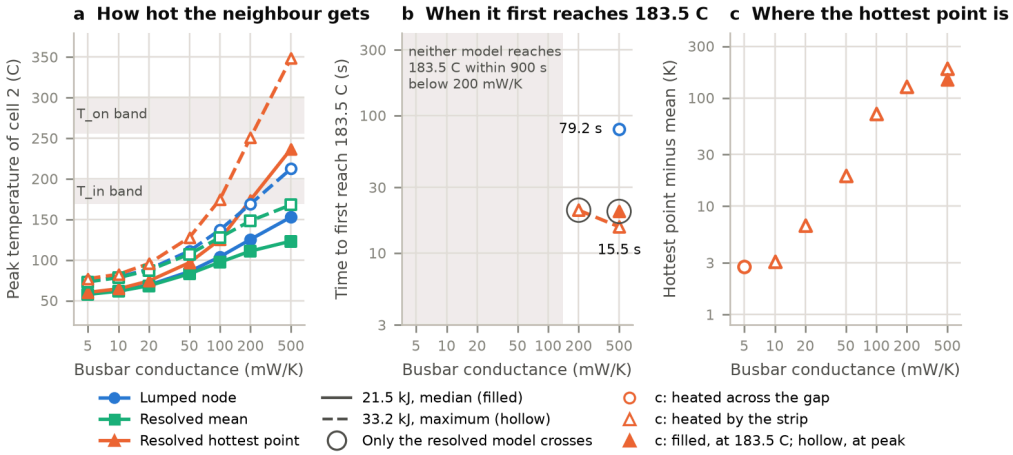


Figure 1. Registered runs, two strips, 30 s release. (a) Peak temperatures against the two-sigma bands of T_in and T_on and (b) first time each model reaches 183.5 C, at the median (filled) and maximum (hollow) energy. (c) Hottest point minus mean at the median energy, at its peak (hollow) and on reaching 183.5 C (filled); triangles are heated mainly by a strip, circles across the gap.

Method

1. Cell 1 is a lumped LG 21700-M50 (bottom-vent variant) releasing its Battery Failure Databank cell-body energy, the minimum, median or maximum of 17 fully charged tests (8.75, 21.5 or 33.2 kJ), as a triangular pulse over 10 or 30 s.
2. Paths to cell 2: two busbars totalling 5 to 500 mW/K (two 0.2 by 8 mm nickel strips 23.1 mm long give 12 mW/K, two 15 mm^2 copper plates 500 mW/K), a 2 mm air gap (17 mW/K) and grey radiation (view factor 0.16); both lose heat to a 25 C room.
3. Cell 2 is solved as one temperature and as a 3D scikit-fem field (roll 1.3 W/mK across and 25 W/mK along the layers, 0.25 mm steel can, strips on an 8 mm cap disc and a rim annulus) with the same cell 1 history, heat capacity, paths and losses.
4. Levels from Koenig, Zhao and Deng (2025), whose simulated ARC test of a 720 g cell with NCM811 kinetics gives main-cluster means T_in = 183.5 C (exotherm trigger at 0.15 C/min) and T_on = 278.9 C (internal short), with two-sigma bands read from their Fig. 7 as 169.0 to 198.0 C and 255.7 to 302.1 C.
5. Checks: heated by radiation alone, the resolved mean stays within 0.98 % of the lumped rise; energy closes to 2 parts in 10^{11} in all 49 sweep runs; at 33.2 kJ and 500 mW/K, halving the time step moves the hottest-point crossings by at most 0.01 s, and halving every mesh spacing by at most 0.09 s.
6. Runs: the 42 cases with two strips, and 7 with one strip at the negative end, as in a series string; at 500 mW/K one strip moves the hottest point to the bottom rim and delays 183.5 C from 20.1 s to 25.0 s (median energy). Exploratory: an isotropic 25 W/mK roll lowers the headline hot spot from 348 C to 248 C, below the T_on band, and delays 183.5 C to 19.3 s.

Limits

1. Conduction and radiation only: ejecta, flame and venting are excluded, so nothing here says whether a real module propagates. Coupling is one-way; in four cases rerun two-way, cell 1 runs hotter, the hottest point reaches 183.5 C at most 0.3 s earlier, and the resolved mean peaks 6 to 14 K higher, at 181 and 182 C for 33.2 kJ and 500 mW/K, just short of 183.5 C.
2. One geometry, with the strips placed directly on the jelly-roll ends; the cap assembly, tabs and headspace are not resolved, which probably exaggerates the terminal hot spot, and the properties are constant.
3. The release is a Databank energy shaped as a triangle, not a measured heat-rate curve; T_in and T_on are whole-cell ARC metrics of a different cell, applied here to a point, to the hottest 1 % and to the mean.

Prediction scorecard

	Prediction	Result	Measured
P1	At 5 mW/K neither model reaches 169.0 C	Held	Peaks of 39 to 77 C
P2	At 500 mW/K and 33.2 kJ the resolved hottest point reaches 183.5 C more than 50 % earlier than the lumped node	Held	92 % and 80 % (central estimate 93 % and 81 %)
P3	Only-resolved 183.5 C crossings begin at 100 or 200 mW/K; at 500 mW/K and 33.2 kJ the hottest point reaches 255.7 C	Held	From 200 mW/K; 255.7 C at 7.1 s and 19.2 s
P4	Hottest point gap-facing at 5 and 10 mW/K, at a terminal end from 100 mW/K	Failed	Switch between 5 and 10 mW/K, where the hottest point is only 3 K above the mean
P5	At 500 mW/K the resolved mean rises at least 15 % less than the lumped node	Held	22 to 24 % less (central estimate 35 to 55 %, too high); two-way: 14.7 to 16.1 % in three cases, one under 15 %

What it means for module-scale modelling

Whether a resolved neighbour reaches onset before or after a lumped one depends on how much of it must be hot: where the lumped node gets there at all (33.2 kJ, 500 mW/K), the hottest point is 64 s sooner and the hottest 1 % 54 to 58 s sooner, but the mean never gets there. Whether onset in a neighbour begins locally is therefore a question a whole-cell threshold cannot settle, so the next step is to run onset kinetics, such as those of Koenig, Zhao and Deng, on a resolved field with the cap and tabs.

Credit: The Battery Failure Databank is provided by the U.S. Department of Energy's National Renewable Energy Laboratory, operated by Alliance for Sustainable Energy, LLC (DOE/NREL/ALLIANCE); its notice accompanies the rows used, in inputs/databank/.

Addendum (v1.1)

1. Yao and Battiato's adaptive hybrid for thermal runaway in a pack (arXiv:2504.10119) couples cells only through a packing continuum, scales temperature by 240 K (their Table 3 and Eq. 6) and keeps its error below the upscaling error of their test cases, 0.05. The results below come from the archived runs, computed after their predictions were public (commit 7b866bf, 12 October 2026, 11:01 UTC); nothing above this heading changed. Version v1.1.0: doi.org/10.5281/zenodo.23315924.
2. Busbars carry 86 to 90 % of the resolved neighbour's heat intake over 900 s at 500 mW/K and still 58 to 66 % at 50 mW/K (two strips, every energy and duration); up to the moment its hottest node first reaches 183.5 C they carry 93.7 and 93.4 % at 500 mW/K and 84.7 and 85.5 % at 200 mW/K (33.2 kJ, 10 and 30 s).
3. In their measure the lumped node's error against the resolved mean peaks at 0.24 at 33.2 kJ and 500 mW/K, but passes 0.05 only at 14.0 and 23.8 s, after the hottest node has reached 183.5 C (5.7 and 15.5 s); at 100 mW/K it stays below 0.05 (peaks 0.043 and 0.044), while at the hottest node, where chemistry starts, the error reaches 0.53 and 0.52.
4. For a two-by-four module of these cells in air at 23.1 mm pitch (ε = 0.25), their packing Biot number is 46, the pack length over the 2 mm gap, and their conductivity ratio 43 (across the layers): both violate their Eq. 13 by those factors before any cell burns, and the burn number, 5.4×10^4 against $1/\varepsilon = 4$, violates it by 1.3×10^4 once one does.
5. In their units a busbar of 50 to 500 mW/K is a coupling Biot number of 130 to 1,300, 3 to 29 times the gap's, carried by a discrete path that their packing continuum does not contain; the internal Biot number of a strip patch, its coefficient times the cell radius over 1.3 W/mK, passes 1 at 12 mW/K on the cap disc and 43 mW/K on the bottom annulus.
6. Scorecard: P1 failed (over 80 % at 500 mW/K as predicted, but 58 to 66 % at 50 mW/K against a predicted under 50 %); P2 failed (peak 0.24 against a central 0.08, but first above 0.05 only after the hottest node's 183.5 C); P3 held (0.043 and 0.044 against a central 0.03); P4 held in part (46 against a central 40, but the conductivity ratio is violated too).

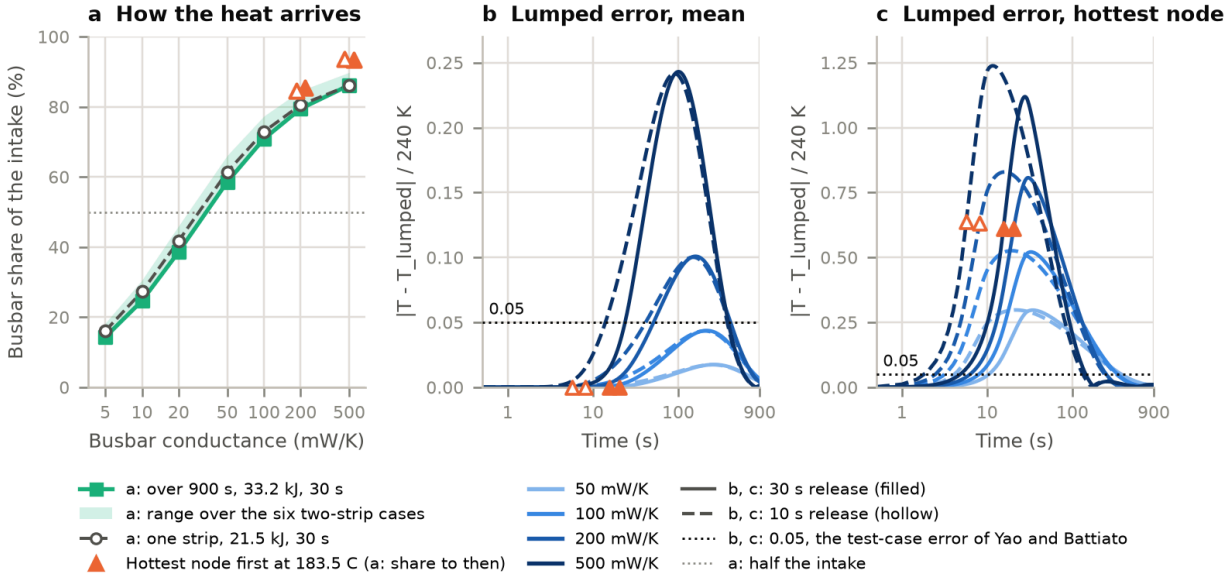


Figure 2. Addendum, registered runs. (a) Busbar share of the resolved neighbour's heat intake over 900 s at 33.2 kJ and 30 s (band: all six two-strip cases at each conductance), up to its hottest node's first 183.5 C (triangles: filled 30 s, hollow 10 s), and with one strip at the negative end (circles). (b, c) The lumped node's error in Yao and Battiato's measure against the resolved mean and against the resolved hottest node, 33.2 kJ, 50 to 500 mW/K (light to dark), 30 s release (solid) and 10 s (dashed), with their test cases' 0.05; triangles mark the hottest node's first 183.5 C.